
Chapter 7: OOD Detection via Crease Ambiguity

(Experiment 4)

A Technical Report from the Puno Calculus Research Program

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Question

Do out-of-distribution inputs produce higher crease density than in-distribution inputs?

The hypothesis: OOD inputs land near many ReLU creases because the network has not learned decisive activation patterns for them. A unit near its crease is maximally uncertain — and a high density of such units signals that the input is unlike anything in the training distribution.

Setup

- **ID data:** Multi-scale checkerboard (5000 training points)
- **Architecture:** MLP [2, 64, 64, 1], 300 epochs
- **Four OOD types:**

OOD Type	Description	Difficulty
Far uniform	Uniform ± 10	Easy (far from data)
Far Gaussian	$N(15, 3)$	Easy (far in space)
Near-OOD	Shifted checkerboard phase	Hard (same support)
Center noise	Uniform ± 1 in center	Hard (within ID region)

• **Metrics:** Per-input crease density (fraction of near-threshold ReLU units), Maximum Softmax Probability (MSP) baseline, AUROC for each method

Results

OOD Type	Crease AUROC	MSP AUROC	Winner
Far uniform (± 10)	0.6085 (ID $\uparrow$)	0.7707 (OOD $\downarrow$)	MSP
Far Gaussian (N(15,3))	0.6798 (ID $\uparrow$)	0.9868 (OOD $\downarrow$)	MSP
Near-OOD (shifted)	0.5114 (ID $\uparrow$)	0.5047 (OOD $\downarrow$)	Tie (chance)
Center noise (± 1)	**0.8835** (OOD $\uparrow$)	0.7061 (ID $\uparrow$)	**Crease**

Key Findings

1. **Crease density and MSP capture complementary failure modes.** This is the most important finding: they are strong in different OOD regimes.
2. **MSP dominates for far-OOD.** The network is overconfident on far-away points, assigning high softmax probabilities to everything. When it encounters truly far-OOD data, this overconfidence collapses, making low MSP a strong far-OOD signal (AUROC up to 0.987).
3. **Crease density dominates for near-boundary ambiguous OOD.** Center noise falls in the region with the most complex boundary structure (the fine checkerboard in the center). Here, crease density achieves AUROC 0.884 — meaning OOD inputs trigger many more near-threshold units than ID inputs.
4. **Both methods fail on near-OOD.** When OOD is structurally similar to ID (same support, shifted phase), both crease density and MSP are near chance (~ 0.5). Near-OOD detection requires more sophisticated methods.
5. **Far-OOD has LOWER crease density than ID**, contradicting the original hypothesis. This confirms the Hein et al. (2019) theoretical result: ReLU networks partition space into unbounded polytopes. Far-away points can fall deep inside a single broad polytope with few crease boundaries nearby. The hypothesis "OOD $\rightarrow$ more creases" holds only for *ambiguous* OOD near the data manifold's complex regions.

Combined Detector

The complementary strengths suggest a practical recommendation: a **combined OOD detector** that uses both signals:

$$OOD_score = \alpha \cdot Energy(x) + (1-\alpha) \cdot CreaseDensity(x)$$

Where α is tuned on a validation set. The energy score (from Liu et al., 2020) is preferred over MSP for its theoretical grounding, but MSP works similarly in practice.

Literature Connection

The closest existing work is the Binary NAP method (Olber et al., CVPR 2023), which uses ReLU on/off binary patterns for OOD detection. The Puno Calculus approach differs in two ways: 1. It counts *near-threshold* units rather than binary on/off states — a continuous signal rather than a discrete one 2. It is grounded in the geometry of ReLU crease boundaries rather than pattern matching

The crease density OOD score is novel. No existing paper proposes counting near-threshold ReLU units as an OOD signal.